Options Selling strategy using Machine Learning

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Abstract

The goal of this paper is to develop a dynamic standalone option selling strategy using technical indicators, options Greeks and machine learning optimization. Options trading is hugely attractive due to the inherent leverage available to take view on market direction. We primarily focus on the options selling side, whose risk profile is akin to selling insurance. Options Sellers typically try to earn the "time value" of the options, i.e. the decay of option's prices with time. We base our options selling strategy on the structure: "Strangle" . To optimize the strategy for the market trends and reversions, we make use of technical indicators, which have been shown to have significant predictive power with regards to market returns. Furthermore, we use options Greeks to take into account the risks associated with the options selling structure. A machine learning model is used to come up with an improved options selling strategy over the vanilla options selling strategy. The options selling strategy outperforms the baseline options selling strategy in both Sharpe ratio and monthly returns. In addition to the strategy implemented, this work provides a robust framework to develop robust machine learning optimized options trading strategies

Keywords: *Options selling, trading strategies, machine learning, technical indicators, option Greeks, call, put, strangle strategy.*

1. Introduction

There are three main types of participants in the options markets: options buyers, options sellers and options market makers. Options Market Makers typically earn the bid-ask spread in the options contract and help connect Options Buyers and Sellers. Option Buyers Consists of participants buying options to hedge their positional trades from hugely adverse moves and participants who take a leveraged position based on an anticipated directional move. Options Sellers, in contrast, have very different objectives: Maximize the premium earned while minimizing the loss arising from adverse moves. Empirically, 80% of the options end up out-of-the-money, putting the odds in favor of the options sellers. But the 20% can entail huge losses, hence Options Selling strategies needs to be executed carefully. Moreover, in many markets, options can be sold with Margin derived from collateral, making it hugely attractive to investors.

The two primary options instruments that are available to trade are the Call Option and the Put option. Buying a Call Option entails a bullish position, so does selling a Put Option. Buying a Put Option entails a bearish position, so does selling a Call Option. The Payoffs of Calls and Put Options (from sellers Perspective are given in Figure 1.

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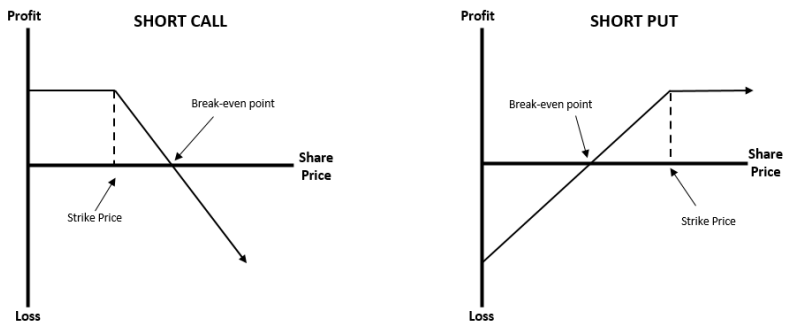

Figure 1 – Short Call and Short Put

Options Sellers use various combinations of these short Call and Put options at various strike prices to earn the premium while minimizing the losses from adverse underlying movement. There are various "structures" - combinations of calls and puts that are used by market participants: Straddles, Strangles, Butterfly, Iron Condors, etc.

In this paper, we use the "Strangle" as the base options selling strategy. Using Machine learning optimization, we aim to maximize the premium earned while minimizing the losses by factoring in anticipated market movement via technical indicators while considering the risk associated with the strangle via the options Greeks. By including technical indicators, we aim to bias/skew the options selling to the direction of the expected market movement by the technical indicators and hence improve the strategy returns. By using options Greek as additional inputs to the model, we include the risks associated with the structure in the model itself and expect a much-improved options selling strategy.

2. Literature Review

Given our aim of developing technical indicators, options Greeks, machine learning optimized options selling strategy, we explore literature and works done in these specific areas. There has been a lot of work in these fields, especially the use of machine learning methods in financial problem statements.

Technical indicators are often used by traders to take directional trades in capital markets. Though contrary to the notion that technical indicators might not be significant predictors due to sole usage by only retail investors, they have been shown to have predictive power. Christopher, David, Jun and Guofu [1] show that technical indicators possess important information which is as significant as macroeconomic factors to model returns, the model used here is the linear regression, the results can possibly be improved by considering more advanced predictive machine learning models. Zhifeng, Xiaodi, Jie and Lianying[2] show the improvement associated with feature engineering: processing the dependent variable to fetch a better predictability. These results might be improved with usage of predictive machine learning models. Zhai, Hsu and Halgamuge [3] show that Support Vector Machines can be fitted on top of technical indicators along with encoded news to trade profitably in capital markets, however they do not consider "shorting" the market on sell signals, limiting the scope.

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Oncharoen and Vateekul [4] employ a deep learning-based predictor on top of technical indicators and news to improve over baseline models. It is important to note that even though deep learning might pick up intricate patterns not picked up by simpler machine learning models, they are also prone to overfitting.

Options trading is considered widely attractive due to the leverage and subsequent high returns associated with it. As is expected, high returns are also accompanied by high risks and it is important to understand all the risks associated with options trading. In the mathematical framework, these risks are represented by the options' Greeks, i.e. sensitivity of options prices to factors like underlying price movement(delta), volatility (vega) and theta (sensitivity to time to expiry).

Options Sellers typically have positive theta and negative vega, i.e. they earn premium over time to expiry and expect volatility to be low for their sold options positions to be profitable. They might or might not have delta. Lai and Kin[5] explore a basic straddle selling strategy and find that it outperforms the index. This is primarily because the implied volatility of the quoted options is often higher than expected movement of the underlying, a spread essentially. The strategy does not consider any dynamic decision factors for if to enter the position, time of entering the position(to expiry) and if to close position based on the profit and loss of the position, these frameworks can be added to improve the return of the strategy. Zabcic-Matic and Tomislav[6] carry out a deep learning based long options strategy based on technical indicators, showing the profitability of machine learning based technical indicator systems. It is important to note that options Greeks are not used in this study and the options prices data is generated rather than using real life options data. The strategy also holds the structure till expiry; hence this is a potential improvement as well.

Constantine[7] shows the optimization of Broken Wing Butterfly options selling strategy using Differential evolution algorithm. Even though the delta of the legs of the butterfly is considered along with time to expiry, implied volatility is not considered. Another potential improvement is the strategy can be further optimally skewed using Technical Indicators. Wen, Yuan and Yang[8] explore a reinforcement learning approach to predict and trade options. Despite Good results, this approach yields the trade-off of loss of explain ability of the trading strategy, i.e. of reinforcement learning agent's actions.

Various ML based models can help predicting the movement of options with various features from the technical analysis data. The automated trading eliminates the uncertainty and the human element in order to automate the options investing decisions. By combining Q-learning, a type of reinforcement learning algorithm, with Black-Scholes model, the conventional option pricing model, a Black-Scholes Q-learning model (QLBS) for determining optimal options prices can be built.[9]

All the areas: Technical Indicators, Machine learning, Options trading and specifically options selling using Greeks and dynamic trading strategy have been studied. But there remains a lack of options selling framework using machine learning optimized technical indicators, Greeks and dynamic trading strategy using real world options data. This is an interesting avenue as options selling is already a widely adopted strategy by sophisticated institutional investors.

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The Base of the Options selling strategy is the Strangle structure. We will optimize for the returns of the short strangle using technical indicators and options Greeks.

A Strangle is simply: Selling an OTM Call Option of certain strike (S1) and Selling an OTM Put option of certain Strike (S2). As long as the underlying price remains between these strikes, the strangle seller earns the time value of the option premium associated with the strikes. The Payoff is explained by Figure 2.

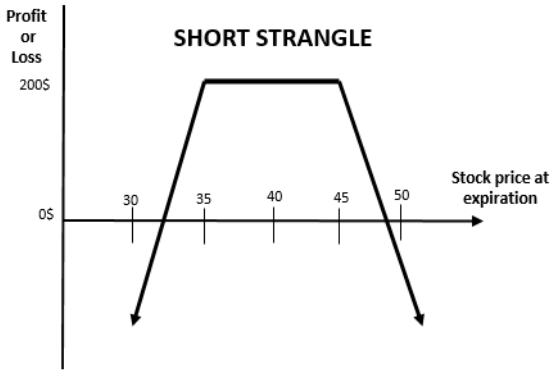

Figure 2 – Short Strangle

The two main reasons for selecting Strangle are:

- **Simplicity**: Strangles are easy to get into and simpler to manage compared to some more complex options selling strategies like Butterflies.
- **Directional Bias**: Strangle allows for a directional Bias as the legs of the structure can be chosen so as to obtain a directional bias.

As is evident, technical indicators can be used to bias the aggressiveness of each leg. If we expect an upside move, we might sell a higher strike Call and higher strike Put as well. The other side of the strategy return: what price is high enough to sell the option and other options risk characteristics will be included in the options Greeks: implied volatility, delta, gamma, theta and time to expiry.

We will use the following option Greeks for a Call/Put option modelling:

- **Implied Vol** is the expected volatility of the underlying stock during the life of an option
- **Delta** measures how much the price of an option would change if the price of its underlying shares changed by one dollar.
- **Theta** indicates the rate of change of option price with respect to decrease in time to expiry(time decay)
- **Gamma** indicates the rate of change of an option's Delta with respect to underlying stock price.

So essentially, for a given straddle (defined by S1 and S2), the inputs to the machine learning model will be: Technical Indicators, options Greeks of the straddle and time to expiry. The Output will be the Expected Return of the strangle if held till expiry. The Training dataset Reward Function will be taken as Expected Return of the strangle if held till expiry. We will sell the strangle if the predicted expected return is more than a threshold (Th).

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3. Methodology

The entire project is done using python. The necessary historical data is downloaded for a chosen option. We plan to use NIFTY(Indian national stock exchange's index option) as it is highly volatile, and data is available.

We create all possible and valid strangle combinations (Call Strike S1, put strike S2) for a given expiry. Valid strangles are determined by Call and Put options both been Out-Of-Money at the time of trade entry(Sell Strangle).

We calculate the margin required for holding the strangle combinations. This is essential to robustly calculate the rate of return on the deployed margin (capital).

We develop a back-tester which outputs the performance metrics: monthly returns and Sharpe ratio by taking in the Sell Signals as the inputs.

For the option legs: i.e. Call and Put option, we calculate the Greek values using python. Furthermore, technical indicators are calculated over the Futures Price. The values are constructed for each historical time period.

The following technical indicators are used in the model:

- **RSI -** Relative Strength Index. Measures rate of price changes.
- **MACD -** Moving Average convergence Divergence. Calculated by subtracting 26-day EMA from 12-day EMA.
- **TRIX -** 1-day Rate-Of-Change (ROC) of a Triple Smooth EMA
- **MOM-** Momentum
- **Rate of Change** (ROC 1, 5 and 10 days)

The following dataset is prepared and filtered as per strangle combinations.

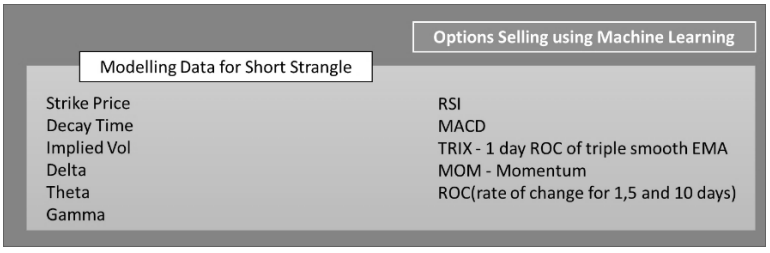

Figure 3 – Short Strangle

For training of ML models, for each strangle combination, we calculate the expected monthly return for each time period. This shall serve as the Target variable for the training of ML model. The ML model is trained and then used to predict expected returns on test dataset.

Below is the summary of the steps:

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1. Collect the options and futures dataset with OHLC prices within a considered timeframe.
2. Create all possible valid strangle combinations.
3. Calculate margin for strangles.
4. Develop a Signal Based Back-tester.
5. Calculate Greeks for the Call and Put legs of the strangles.
6. Calculate Technical Indicators Based on Futures Price History.
7. Create Train and Test Dataset.
8. Calculate forward looking expected strangle return for train dataset
9. Train ML Model.
10. Test the following strategy on the test dataset: Sell the straddle if Expected Strangle Return > Threshold (Th), with preset Stop Loss and Take Profit (exit) Conditions.
11. Compare against Baseline Strangle Selling strategy: Sell Strangle if Put and Call are 5% Out of the Money and Days to expiry is more than 20.

The design of the application is shown below. We plan to use Non-Linear ML Models (Decision Trees, etc.) to give the best classification model for the training data taken for building the model.

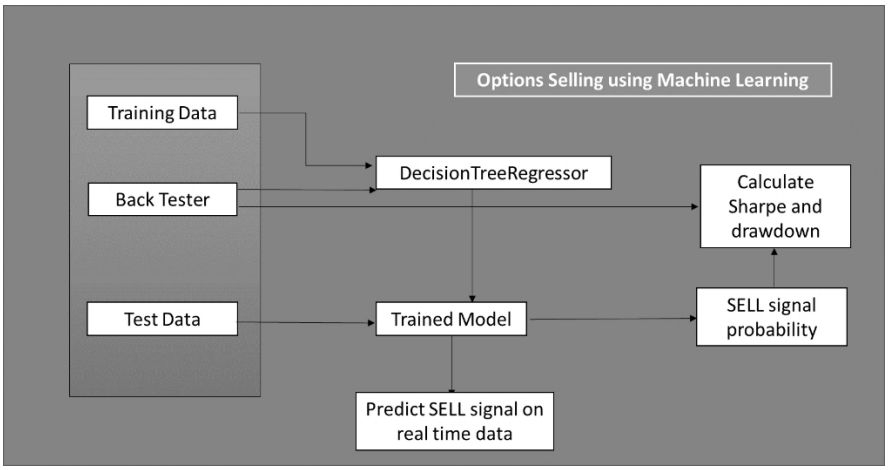

Figure 4 - System Design

4. Source Code structure

The coding is done in Python 3.7. Following is the set of modules in the code :

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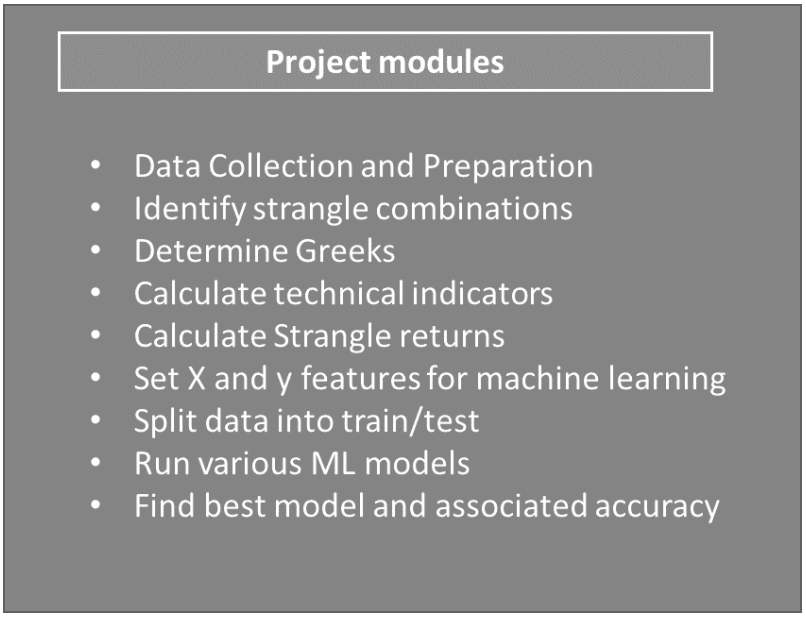

Figure 5 – Project modules

Each of the module mentioned perform their own processing and help achieve the final goal of setting up a trading system of options selling using machine learning.

The modules are explained in detail in 4.1

4.1. Data collection and preparation

For training and testing the machine learning model, we gather the options data for NIFTY and setup other features of technical indicators and option Greeks. The historical data is extracted from NSE(National Stock Exchange of India) website for the past 5 years – 2017 to 2021. For each year and 12 months within a year, the weekly expiry date is first extracted.

Then, for each expiry date, the following columns are extracted:

- Symbol – the underlying stock symbol
- Expiry – date of expiry(Thursday of every week)
- Option Type(Future/PE/CE)
- Strike Price – strike price of the option
- Open – opening price of the day
- High – day's high
- Low – day's low
- Close – closing price of the day
- Last – Last traded price.

The data is extracted for each trading day of each year. Hence, for each trading date, for each expiry date, for each strike price, there are 3 sets of rows. Sample data below:

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Date	Symbol	Expiry	Option Type	Strike Price	Open	High	Low	Close	Last
2017-01-02	NIFTY	2017-01-25	FUT	NaN	8215.05	8218.8	8151.00	8192.10	8191.10
2017-01-03	NIFTY	2017-01-25	FUT	NaN	8200.00	8228.0	8159.00	8196.50	8189.55
2017-01-04	NIFTY	2017-01-25	FUT	NaN	8202.25	8223.4	8187.30	8204.05	8207.90
2017-01-05	NIFTY	2017-01-25	FUT	NaN	8254.70	8298.7	8236.15	8288.20	8287.50
2017-01-06	NIFTY	2017-01-25	FUT	NaN	8305.55	8329.6	8256.50	8267.65	8268.10
...	...	...	...	...	...	...	...	...	...
2021-12-24	NIFTY	2021-12-30	PE	19400.0	2406.25	2452.5	2406.25	2448.00	2448.00
2021-12-27	NIFTY	2021-12-30	PE	19400.0	0.00	0.0	0.00	2448.00	2448.00
2021-12-28	NIFTY	2021-12-30	PE	19400.0	0.00	0.0	0.00	2448.00	2448.00
2021-12-29	NIFTY	2021-12-30	PE	19400.0	0.00	0.0	0.00	2448.00	2448.00
2021-12-30	NIFTY	2021-12-30	PE	19400.0	0.00	0.0	0.00	2448.00	2448.00

As much data which is extractable from the NSE site is taken. Exceptions are handled accordingly. The following NSE packages are used:

- nsepy.derivatives - get_expiry_date
- nsepy - get_history

4.2. Creating strangle combinations

From the extracted data for 67046 rows, we need to find strangle combinations for our model. Here, we take the unique future prices and find OTM PE and CE options for each expiry week and each trading date. This data is merged and set of option strangles are obtained. The PE and CE data are obtained in the same row.

Sample screenshot below:

Date	Symbolleg1	Expiry	Option Typeleg1	Strike Priceleg1	Openleg1	Highleg1	Lowleg1	Closeleg1	Lastleg1	Close_FUTleg1	Symbolleg2	T
2017-01-02	NIFTY	2017-01-25	CE	7300.0	889.15	901.95	860.7	901.95	901.95	8192.10	NIFTY	
2017-01-03	NIFTY	2017-01-25	CE	7300.0	886.60	919.00	858.0	887.70	884.50	8196.50	NIFTY	
2017-01-04	NIFTY	2017-01-25	CE	7300.0	895.55	919.80	889.0	895.95	896.00	8204.05	NIFTY	
2017-01-05	NIFTY	2017-01-25	CE	7300.0	915.90	990.25	915.7	976.45	974.95	8288.20	NIFTY	
2017-01-06	NIFTY	2017-01-25	CE	7300.0	1000.50	1020.00	953.0	960.30	958.20	8267.65	NIFTY	
...	...	...	...	...	...	...	...	...	...	...	...	
2021-12-24	NIFTY	2021-12-30	CE	15300.0	0.00	0.00	0.0	1332.65	0.00	17003.80	NIFTY	
2021-12-27	NIFTY	2021-12-30	CE	15300.0	0.00	0.00	0.0	1332.65	0.00	17103.40	NIFTY	
2021-12-28	NIFTY	2021-12-30	CE	15300.0	0.00	0.00	0.0	1332.65	0.00	17246.75	NIFTY	
2021-12-29	NIFTY	2021-12-30	CE	15300.0	1945.50	1945.50	1945.5	1945.50	1945.50	17215.90	NIFTY	
2021-12-30	NIFTY	2021-12-30	CE	15300.0	0.00	0.00	0.0	1945.50	1945.50	17201.70	NIFTY	

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Option Typeleg2	Strike Priceleg2	Openleg2	Highleg2	Lowleg2	Closeleg2	Lastleg2	Close_FUTleg2
PE	7300.0	3.50	4.80	3.40	4.10	4.20	8192.10
PE	7300.0	3.95	4.50	2.85	3.25	3.35	8196.50
PE	7300.0	3.00	3.30	2.65	2.95	2.85	8204.05
PE	7300.0	1.40	2.85	1.40	2.00	1.95	8288.20
PE	7300.0	1.80	1.95	1.45	1.60	1.45	8267.65
---	---	---	---	---	---	---	---
PE	15300.0	3.10	3.10	1.75	1.95	1.90	17003.80
PE	15300.0	2.20	2.20	1.60	1.80	1.85	17103.40
PE	15300.0	1.35	1.65	1.05	1.25	1.25	17246.75
PE	15300.0	1.25	1.25	0.75	0.85	0.80	17215.90
PE	15300.0	0.75	0.75	0.05	0.15	0.05	17201.70

From the entire set, filtering is done to get strangle options which are OTM. The margin is also calculated for each strangle combination.

4.3. Determining Greeks

With the dataset obtained for valid strangle combinations, we drop rows with any NULL values. The cleaned dataset is then used to build features. The following option Greeks are added as new features for each row's PE and CE.

- Call Delta
- Call Vega
- Call Theta
- Call Gamma
- Put Delta
- Put Vega
- Put Theta
- Put Gamma

The python package mibian is used to calculate the Greeks. The functions called are:

- mibian.BS.callDelta
- mibian.BS.vega
- mibian.BS.callTheta
- mibian.BS.gamma

4.4. Calculating technical indicators

New features with technical indicators are added for each strangle combination row. The following technical indicators are added, and the associated library used is mentioned:

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- ADX - talib.ADX(High, Low, Close, timeperiod=14)
- RSI - talib.RSI(Close, timeperiod=14)
- MACD - talib.MACD(Close, fastperiod=12, slowperiod=26, signalperiod=9)
- TRIX - talib.TRIX(Close, timeperiod=30)
- MOM - talib.MOM(Close, timeperiod=10)
- ROC_10 - talib.ROC(Close, timeperiod=10)
- ROC_5 - talib.ROC(Close, timeperiod=5)
- ROC_1 - talib.ROC(Close, timeperiod=1)

4.5. Calculating Strangle returns

The target feature with the built dataset will help us build the model. The target feature is a SELL signal based on return on the strangle. The return on each strangle is calculated as:

Strangle_Return=(Strangle_Price-StranglesMonthClosePrice)*100/strangles_margin

More the return, better is the opportunity to SELL.

The final prepared dataset looks like below:

Date	Symbolleg1	Expiry	Option Typeleg1	Strike Priceleg1	Openleg1	Highleg1	Lowleg1	Closeleg1	Lastleg1	...	MACD	TRIX	MOM	ROC_10	ROC_5
2017-01-02	NIFTY	2017-01-25	CE	7300.0	889.15	901.95	860.70	901.95	901.95	...	NaN	NaN	NaN	NaN	NaN
2017-01-03	NIFTY	2017-01-25	CE	7300.0	886.60	919.00	858.00	887.70	884.50	...	NaN	NaN	NaN	NaN	NaN
2017-01-04	NIFTY	2017-01-25	CE	7300.0	895.55	919.80	889.00	895.95	896.00	...	NaN	NaN	NaN	NaN	NaN
2017-01-05	NIFTY	2017-01-25	CE	7300.0	915.90	990.25	915.70	976.45	974.95	...	NaN	NaN	NaN	NaN	NaN
2017-01-06	NIFTY	2017-01-25	CE	7300.0	1000.50	1020.00	953.00	960.30	958.20	...	NaN	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2021-12-24	NIFTY	2021-12-30	CE	19400.0	1.30	1.60	0.90	1.20	0.90	...	-209.731601	0.076876	-555.55	-3.163841	-0.115722
2021-12-27	NIFTY	2021-12-30	CE	19400.0	1.00	1.05	0.55	0.75	0.60	...	-196.417140	0.068146	-343.45	-1.968550	2.815441
2021-12-28	NIFTY	2021-12-30	CE	19400.0	0.75	0.75	0.50	0.70	0.60	...	-172.311882	0.060073	-108.30	-0.624026	2.653421
2021-12-29	NIFTY	2021-12-30	CE	19400.0	0.75	0.75	0.30	0.40	0.35	...	-153.923297	0.052570	-44.90	-0.260127	1.429005
2021-12-30	NIFTY	2021-12-30	CE	19400.0	0.25	0.60	0.05	0.10	0.05	...	-138.894938	0.045587	-115.80	-0.668688	0.721670

4.6. Setting X and y features for modelling.

The following features are set as X and y:

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X columns:

- Strangle_Price
- CE_ITMNESS
- PE_ITMNESS
- Days_To_Expiry
- CallIV
- CallDelta
- CallVega
- CallTheta
- CallGamma
- PutIV
- PutDelta
- PutVega
- PutTheta
- PutGamma
- ADX
- RSI
- MACD
- TRIX
- MOM
- ROC_10
- ROC_5
- ROC_1

y column:

- Strangle_Return

4.7. Train/test split and modelling

We have setup the data for regression modelling. The objective is to choose a best model to predict strangle returns with the given historical data. The data is split to 80:20 for training and testing.

The DecisionTreeRegressor model is applied on the data for predicting whether a particular strangle combination is worth selling.

4.8. Deliverables

The deliverables are:

- Python code with all modules compatible with version 3.7
- Requirements.txt for packages to be installed

The entire code base is stored in github as read-only.

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5. Discussion and Results

From the back-test results of the ML optimized strategy versus that of the baseline strangle selling strategy, it is clear that inclusion of options Greeks (risk factors), technical indicators(RSI,MACD,MOM,ROC) in ML model to optimize for strangle selling decisions significantly increases the investment returns.

The framework that is developed takes all the information present in the market: options Greeks, technical indicators, compares it to past scenarios (training of ML model) and comes up with the selling decision, all in an automated manner. Automation is particularly important and gives an edge over human generated trading strategy(baseline) in two important ways :

- Exploiting consistent high dimensional patterns in markets that may not be visible to human traders
- Letting go of any human bias. These two factors are where the edge of a statistics/ML based automated trading system lies.

The ML optimized strangle selling strategy outperforms the baseline strangle selling strategy on the test dataset by 22% on an annualized return basis. We include a threshold parameter (Th), which the investor can use as a confidence parameter to take more risk for obtaining more return. In the back test, we have set this value as 6% expected monthly return from the model to sell the strangle(Entry Condition). The exit conditions are either of these: SL hit, TP hit, Closed at Last day of expiry.

The performance is also extended to the amount of risk taken to generate the return (given by Sharpe ratio and max drawdown), which is lower in the ML optimized strategy compared to the baseline strategy. The Sharpe ratio of the ML optimized strategy is 1.33 compared to 0.46 of the baseline strangle selling strategy. The maximum drawdown encountered was also lower for the ML optimized strategy compared to the baseline strategy: -3.4% vs -4.7%.

	Simple Strangle Sell	ML based Strangle SELL
Overall Trades	1144	1088
Overall profits	45812.0000	114600.0000
Overall Margin used	3711141.1000	3671008.1000
Profit Percentage	1.2584	3.1262
Sharpe ratio	0.4657	1.3026

The improvements of the ML optimized strategy can be attributed to 3 factors –

1. Usage of Options Risk factors(Greeks).
2. Usage of Technical Factors.
3. Usage of ML model to uniquely find mispricing in the market combination defined by options risk factors and technical indicators.

The Machine learning model learns the mapping of the market state space defined by (Options Greeks, Technical Indicators) to expected Monthly Strangle return. Usage of non-linear models is preferred as patterns in the market are often non-linear. In the case of this project, the mapping is of the form of a Decision Tree.

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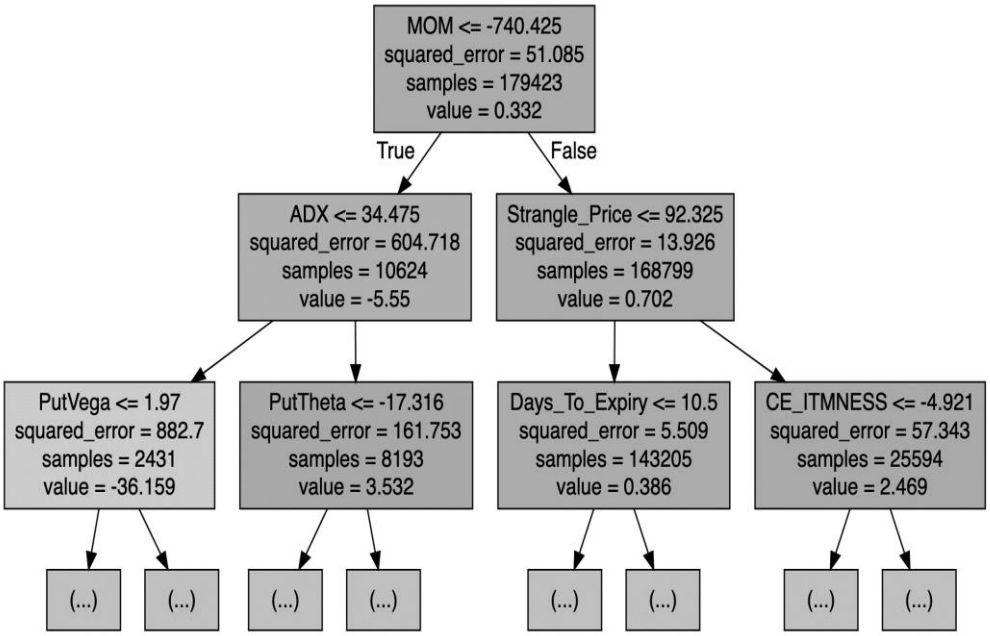

Figure 6 – Project modules

Here we can see the decision path that the model learns from the training data. The generated decision paths can be used by investors to better understand the learnings of the model for the trading strategy. Consider the sample Path: MOM (momentum)>-740.425, Strangle_Price<92.325and Days to Expiry < 10.5, here the expected Monthly Strangle return in 0.386, implying its best not to set Strangle in this market space. Hence, the framework can also be used to generate confidence in the underlying Machine Learning model by comparing actual decision paths with respect to the options risk factors(Greeks) and technical indicators. The Top 10 Factors which are ranked according to their importance in decision making are as follows:

	feature_importance
ADX	0.348203
PutVega	0.191480
PutIV	0.062434
PutTheta	0.054278
MOM	0.044283
MACD	0.041658
Strangle_Price	0.040837
TRIX	0.040061
RSI	0.024471
PutGamma	0.023296

These feature importance gives the investor information about which factors are dominating the decision-making process according to the model which is fit.

The framework can be used with the traditional Take Profit and Stop Loss processes often used by investors. Hence, the distribution of false negatives and false positives are caught in the back test itself, identifying optimal take profit and stop can be further performed.

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6. Conclusions

The option selling strategy has been successfully implemented with machine learning and compared with simple strangle strategy. The model can be taken out as a pickle file and on a real-time basis, current option and strike price data with strangle combination can be used to predict whether the SELL return is good or not. Depending on the risk:reward ratio, the module using this model can be setup for automated trading.

In future, the following can be added to the project:

- Use BANK NIFTY or any other volatile option
- The machine learning model can be made to use AutoML by adding H2O for regression.

The project outcome is a robust option selling strategy and a robust platform on which further additions can be incorporated in the current strategy. The code and platform are production ready.

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Disclaimer

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Appendix

A. Resources

Code repository: *https://github.com/athujoshi/WQU-Options-Selling-ML*

Python: https://www.python.org/

Pandas: https://pandas.pydata.org/

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